RIG-FLATNESS IS STABLE UNDER ADIC TOPOLOGICALLY FINITE TYPE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the author also flags the identity as something used below; it is the compressed standard fact, not a verification specific to the claim

Editorial note. The label lemma-rig-flat-base-change names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Lemma lemma-rig-closed-point-relative (delivered, L48108): contraction behaviour of rig-closed primes through adic topologically finite type maps
- Lemma lemma-rig-closed-point-variables (in-source): at rig-closed primes the completions of the restricted and polynomial power series local rings are isomorphic
- More on Algebra, Lemma more-algebra-lemma-flat-completion: flatness transfers through isomorphisms on completions
- Algebra, Lemma algebra-lemma-completion-tensor and algebra-lemma-flat-base-change
- standard formal-space identity A_{f}{x} = A{x}_{f: localization commutes with restricted power series

Source reference. Original lemma 00MA: sources/stacks-chow/tags.txt, source lines 816–816. Prerequisite pointer invoked at 3819; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00HI: sources/stacks-chow/tags.txt, source lines 644–644. Prerequisite pointer invoked at 3821; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGG: sources/stacks-restricted/pinned-original-source.tex, source lines 3488–3488. Prerequisite pointer invoked at 3842; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGB: sources/stacks-restricted/pinned-original-source.tex, source lines 3333–3333. Prerequisite pointer invoked at 3868; it names

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a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GGH: sources/stacks-restricted/pinned-original-source.tex, source lines 3538–3538. Prerequisite pointer invoked at 3881; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C4G: sources/stacks-chow/tags.txt, source lines 15226–15226. Prerequisite pointer invoked at 3888; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\varphi : A \to B$ and $A \to C$ be arrows of $WAdm^{Noeth}$. Assume φ is rig-flat and $A \to C$ adic and topologically of finite type. Then $C \to B\widehat{\otimes}_A C$ is rig-flat.*

Proof. Assume φ is rig-flat. Let $f \in C$ be an element. We have to show that $C_{\{f\}} \to B\widehat{\otimes}_A C_{\{f\}}$ is naively rig-flat. Since we can replace C by $C_{\{f\}}$ we it suffices to show that $C \to B\widehat{\otimes}_A C$ is naively rig-flat.

If $A \to C$ is surjective or more generally if C is finite as an A-module, then $B \otimes_A C = B\widehat{\otimes}_A C$ as a finite module over a complete Noetherian ring is complete, see Algebra, Lemma 00MA. By the usual base change for flatness (Algebra, Lemma 00HI) we see that naive rig-flatness of φ implies naive rig-flatness for $C \to B \times_A C$ in this case.

In the general case, we can factor $A \to C$ as $A \to A\{x_1, \ldots, x_n\} \to C$ where $A\{x_1, \ldots, x_n\}$ is the restricted power series ring and $A\{x_1, \ldots, x_n\} \to C$ is surjective. Thus it suffices to show $C \to B\widehat{\otimes}_A B$ is naively rig-flat in case $C = A\{x_1, \ldots, x_n\}$. Since $A\{x_1, \ldots, x_n\} = A\{x_1, \ldots, x_{n-1}\}\{x_n\}$ by induction on n we reduce to the case discussed in the next paragraph.

Here $C = A\{x\}$. Note that $B\widehat{\otimes}_A C = B\{x\}$. We have to show that $A\{x\} \to B\{x\}$ is naively rig-flat. Let $\mathfrak{q} \subset B\{x\}$ be a rig-closed prime ideal. We have to show that $B\{x\}_{\mathfrak{q}}$ is flat over $A\{x\}$. Set $\mathfrak{p} = A \cap \mathfrak{q}$. By Lemma 0GGG we can find an $f \in A$ such that f maps to a unit in $B\{x\}/\mathfrak{q}$ and such that the prime ideal $\mathfrak{p}'$ in $A_{\{f\}}$ induced is rig-closed. Below we will use that $A_{\{f\}}\{x\} = A\{x\}_{\{f\}}$ and similarly for B; details omitted. Consider the diagram

$$\begin{array}{ccc} (B\{x\})_{\mathfrak{q}} & \longrightarrow & (B_{\{f\}}\{x\})_{\mathfrak{q}'} \\ \uparrow & & \uparrow \\ A\{x\} & \longrightarrow & A_{\{f\}}\{x\} \end{array}$$

We want to show that the left vertical arrow is flat. The top horizontal arrow is faithfully flat as it is a local homomorphism of local rings and flat as $B_{\{f\}}\{x\}$ is the completion of a localization of the Noetherian ring $B\{x\}$. Similarly the bottom horizontal arrow is flat. Hence it suffices to prove that the right vertical arrow is flat. This reduces us to the case discussed in the next paragraph.

Here $C = A\{x\}$, we have a rig-closed prime ideal $\mathfrak{q} \subset B\{x\}$ such that $\mathfrak{p} = A \cap \mathfrak{q}$ is rig-closed as well. This implies, via Lemma 0GGB, that the intermediate primes

$B \cap \mathfrak{q}$ and $A\{x\} \cap \mathfrak{q}$ are rig-closed as well. Consider the diagram

$$\begin{array}{ccc} (B[x])_{B[x]\cap\mathfrak{q}} & \longrightarrow & (B\{x\})_{\mathfrak{q}} \\ \uparrow & & \uparrow \\ (A[x])_{A[x]\cap\mathfrak{q}} & \longrightarrow & (A\{x\})_{A\{x\}\cap\mathfrak{q}} \end{array}$$

of local homomorphisms of Noetherian local rings. By Lemma 0GGH the horizontal arrows define isomorphisms on completions. We already know that the left vertical arrow is flat (as $A \to B$ is naively rig-flat and hence $A[x] \to B[x]$ is flat away from the closed locus defined by an ideal of definition). Hence we finally conclude by More on Algebra, Lemma 0C4G. □